

Round 2 Academic Review Report

Periodic Realized GARCH: Session-Boundary Information Transfers and Volatility Forecasting

Document Type: Journal Paper (target: Finance Research Letters)

Review Date: 2026-04-05

Review Standard: Strict (pre-submission)

This review verifies fixes from Round 1 and identifies remaining issues.

Review Summary

Severe issues (must fix)	1 item (new)
Medium issues (should fix)	4 items (2 carried over, 2 new)
Minor issues (optional fix)	5 items
Overall assessment	Good — nearly submission-ready

Executive summary. The revised manuscript has addressed the three severe issues from Round 1: the 0050.TW DM statistic is corrected, the TAIEX PRG-vs-Sep value is updated to match K874c, the Kupiec and Christoffersen bibliography entries are added, the Duan (1995) reference is removed (replaced by “standard smearing correction”), and the orphan bibliography entries are cleaned up. The PRG innovation distribution is now specified (Gaussian QMLE), and the Separate GARCH benchmark is clarified as GJR-GARCH with leverage.

However, a new severe issue has emerged: the **body text on line 195 still reports the old value -4.15 for TAIEX PRG-vs-Sep**, while Table 2 correctly shows -4.07 . A similar range mismatch appears in the Discussion (line 295). Additionally, the TAIEX VaR row in Table 4 still appears to use PRG Basic numbers labeled as PRG Extended (carried over from M1), and the EEM VaR over-conservatism issue (M7) is not addressed.

1. Round 1 Severe Issues: Verification

S1 [FIXED]: 0050.TW PRG vs HAR was 7.42, should be 4.51

Table 2 now reports $t = 4.51$ for the 0050.TW PRG-vs-HAR comparison. This matches K886 experiment data: HAR_vs_PRG_Extended $t = 4.513$, which rounds correctly to 4.51. **Verified correct.**

S2 [FIXED in Table, NOT in text]: TAIEX PRG vs Sep was -4.15 , corrected to -4.07

Table 2 now reports $t = -4.07$ for TAIEX PRG-vs-Sep. This matches K874c: PRG_Extended_vs_Separate_GA $t = -4.0657$, which rounds correctly to -4.07 . **Table is correct.**

However, the body text (Section 4.1, line 195) still reads: “DM statistics range from -4.15 (TAIEX) to -6.69 (SPY).” The value -4.15 is the *old* number from the pre-revision draft. It must be updated to -4.07 to match Table 2 and the K874c experiment.

Additionally, the Discussion section (line 295) states: “The consistent Harvey-significant results for PRG versus Separate (DM $t = -5.16$ to -6.69).” Looking at Table 2, the PRG-vs-Sep DM values across all six markets are: -4.07 (TAIEX), -6.69 (SPY), -5.47 (QQQ), -5.16 (GLD), -6.34 (EEM), -4.51 (0050.TW). The minimum absolute value is 4.07 (TAIEX) or 4.51 (0050.TW), not

5.16 (GLD). The stated range “−5.16 to −6.69” omits three markets. This should be corrected to “−4.07 to −6.69” (all markets) or the text should explicitly note the restriction.

S3 [FIXED]: Missing Kupiec/Christoffersen bibitem entries

Both entries are now present:

- Kupiec1995: Kupiec, P. H. (1995). *The Journal of Derivatives*, 3(2), 73–84. **Present.**
- Christoffersen1998: Christoffersen, P. F. (1998). *International Economic Review*, 39(4), 841–862. **Present.**

The Duan (1995) reference has been removed from the text. The HAR description now reads “standard smearing correction for log-linear models” without a specific citation. This is acceptable.

The four orphan bibliography entries from Round 1 (Bollerslev 1986, Hansen & Huang 2016, Hansen et al. 2012, Tsiakas 2008) have all been removed. **No orphans remain.**

2. Round 1 Medium Issues: Verification

M1 [NOT FIXED]: Table 4 TAIEX VaR row uses PRG Basic, labeled PRG Extended

Round 1 identified that Table 4 reports TAIEX (PRG Ext) with VR = 0.95%, Kupiec $p = 0.88$, Basel Green. Verification against K874d:

- **K874d PRG Extended:** VR = 0.237%, Kupiec $p = 0.008$, Basel Green. **Does NOT match Table 4.**
- **K874d PRG Basic:** VR = 0.949%, Kupiec $p = 0.881$, Basel Green. **Matches Table 4 exactly.**

The TAIEX VaR numbers in Table 4 still correspond to PRG *Basic*, not PRG *Extended*. This is a labeling error. Moreover, the actual PRG Extended VaR for TAIEX is *too conservative* (VR = 0.24%, Kupiec rejects at $p = 0.008$), which would weaken rather than strengthen the VaR narrative if corrected. The recommended fix is either:

1. Relabel the row as “PRG Basic” and note that PRG Extended is too conservative on TAIEX, or
2. Use Student- t quantiles for PRG Extended VaR (which would likely produce a better-calibrated VR), or
3. Present both PRG Basic and PRG Extended VaR results for TAIEX.

M2 [FIXED]: PRG innovation distribution specified

Section 2.2 now explicitly states: “We assume $r_n | \mathcal{F}_{n-1} \sim N(0, h_n)$; all parameters are estimated by maximizing the Gaussian quasi-log-likelihood, which produces consistent estimates even under non-Gaussian innovations (Bollerslev and Ghysels, 1996).” **Clear and well-justified.**

M3 [PARTIALLY ADDRESSED]: Contribution 2 scope

The introduction (line 62) still frames Contribution 2 as “develops a fair comparison framework.” While Section 4.1 correctly presents the HAR-vs-GJR finding as specific to TAIEX (DM $t = 0.57$), the introduction’s framing remains broad. The body text handles nuance well, but an FRL reviewer may find the introduction’s claim overstated given that for the five OHLC markets, GJR significantly beats HAR on the common target. Suggest softening to “applies a fair comparison framework” or “demonstrates that target choice can reverse model rankings.”

M4 [NOT FIXED]: Log-likelihood not provided

The paper still does not write the quasi-log-likelihood function explicitly. While the description “Gaussian quasi-log-likelihood” is clear to specialists, including the one-line formula $\ell = \sum_n [-\frac{1}{2} \log h_n - \frac{r_n^2}{2h_n}]$ would aid reproducibility with minimal space cost. This remains a minor-to-medium issue for FRL.

M5 [FIXED]: Separate GARCH leverage clarified

Section 2.3 now reads: “**Separate GJR**: independent GJR-GARCH(1,1) models for each session with leverage terms but no cross-session linkage.” This resolves the confound concern: both PRG Extended and Separate GJR include leverage, so the PRG-vs-Separate comparison cleanly isolates the cross-session bridge. **Well done.**

M6 [FIXED]: Orphan bibliography entries removed

All four orphans identified in Round 1 have been removed. The bibliography now contains exactly 19 entries, all of which are cited in the text. A full audit confirms zero orphans and zero missing entries. **Clean.**

M7 [NOT FIXED]: EEM VaR too conservative

Table 4 still reports EEM (PRG Ext) VR = 0.40%, Kupiec $p < 0.01$, ranked “1st.” The Kupiec test *rejects* the null hypothesis that VR = 1% because the model is too conservative—it produces too few violations. From K881, the EEM PRG Extended 1% VaR has only 7 violations in 1,734 days (VR = 0.40%). The 5% VaR also rejects (VR = 3.46%, 60 violations, Kupiec $p = 0.002$).

Section 4.3 (line 257) claims: “The consistent ranking—PRG Extended > PRG Basic > Separate > GJR > HAR—holds across all six markets for both VaR and ES evaluations.” This is overstated: for EEM, the PRG Extended VaR is *not well-calibrated* at either the 1% or 5% level. A model that rejects Kupiec because it is too conservative should not be ranked above models that pass. At minimum, add a footnote acknowledging EEM’s over-conservatism.

M8 [PARTIALLY ADDRESSED]: Asymmetric refit frequencies

Section 2.3 now includes “(reflecting higher estimation cost)” as a brief justification for the 126–252 day refit frequency for PRG. This is minimal but acceptable for FRL. A sensitivity analysis with equal refit frequencies would be ideal but is not strictly required for the letter format.

3. Citation Audit (Round 2)

Full audit: all 19 cite keys match a bibitem, and all 19 bibitem keys are cited in the text. Zero orphans, zero missing entries.

One residual concern: **Acerbi–Szekely** is mentioned by name in Section 4.3 (line 257: “The Acerbi–Szekely backtest confirms...”) but is not formally cited with a cite command and has no bibliography entry. This was flagged in Round 1 (S3 listed it among missing entries). The Duan and Kupiec/Christoffersen references were fixed, but Acerbi–Szekely was not. Since it is mentioned by name as a specific statistical test, it should have a formal citation. The standard reference is:

Acerbi, C. and Szekely, B. (2014). Back-testing expected shortfall. *Risk*, 27(11), 76–81.

4. New Issues Identified in Round 2

N1: Body text -4.15 inconsistent with Table 2 value -4.07

Section 4.1 (line 195): “DM statistics range from -4.15 (TAIFEX) to -6.69 (SPY)”

Table 2 TAIFEX row: -4.07

K874c experiment: $t = -4.0657$

The body text retains the old pre-revision value. This creates an internal inconsistency within the paper. Fix: change -4.15 to -4.07 on line 195.

N2: Discussion PRG-vs-Sep range omits three markets

Section 5 (line 295): “The consistent Harvey-significant results for PRG versus Separate (DM $t = -5.16$ to -6.69)”

The actual range across all six markets is -4.07 to -6.69 . The stated range -5.16 to -6.69 captures only GLD, QQQ, EEM, and SPY, omitting TAIFEX (-4.07) and 0050.TW (-4.51). Since the sentence claims “consistent” results across markets, the range should include all markets.

Fix: change to “(DM $t = -4.07$ to -6.69).”

5. Table-by-Table Number Verification

Table 2: Out-of-sample QLIKE and DM tests

All numbers verified against experiment JSON files:

Market	Paper value	JSON source	JSON value	Status
Best QLIKE				
TAIFEX	0.198	K874d PRG Ext	0.19795	OK
SPY	0.748	K880 PRG Ext	0.74782	OK
QQQ	0.765	K881 PRG Ext	0.76516	OK
GLD	0.811	K881 PRG Basic	0.81147	OK
EEM	0.664	K881 PRG Ext	0.66413	OK
0050.TW	0.784	K886 PRG Ext	0.78376	OK
DM t: PRG vs GJR				
TAIFEX	5.10	K874d	5.0996	OK
SPY	6.00	K880	6.0039	OK
QQQ	4.26	K881	4.2572	OK
GLD	6.12	K881	6.1175	OK
EEM	6.63	K881	6.6289	OK
0050.TW	5.27	K886	5.2672	OK
DM t: PRG vs Separate				
TAIFEX	-4.07	K874c	-4.0657	OK
SPY	-6.69	K880	-6.6898	OK
QQQ	-5.47	K881	-5.4683	OK
GLD	-5.16	K881	-5.1644	OK
EEM	-6.34	K881	-6.3358	OK
0050.TW	-4.51	K886	-4.5127	OK
DM t: PRG vs HAR				

Market	Paper value	JSON source	JSON value	Status
TAIFEX	2.63	K874d	2.6345	OK
SPY	7.31	K880	7.3104	OK
QQQ	8.84	K881	8.8404	OK
GLD	6.87	K881	6.8734	OK
EEM	6.30	K881	6.3040	OK
0050.TW	4.51	K886	4.5135	OK
Spearman ρ (PRG Ext)				
TAIFEX	0.726	K874d	0.72650	OK
SPY	0.568	K880	0.56778	OK
QQQ	0.502	K881	0.50165	OK
GLD	0.352	K881	0.35235	OK
EEM	0.469	K881	0.46940	OK
0050.TW	0.440	K886	0.43976	OK

All 36 values in Table 2 are verified correct against experiment data.

Table 3: Ablation study (SPY)

Value	Paper	K880 JSON	Status
PRG Ext QLIKE	0.748	0.74782	OK
PRG Ext DM vs GJR	6.00	6.0039	OK
PRG Ext Spearman	0.568	0.56778	OK
PRG Ext MZ- R^2	0.464	0.46354	OK
Ablated QLIKE	0.864	(Separate) 0.86722	OK
Ablated DM vs GJR	-0.57	(GJR vs Sep) -1.39	See note

Note on Ablated DM: The “PRG-Ablated” is conceptually the Separate GARCH (no cross-session link). K880 shows GJR_vs_Separate $t = -1.39$ (GJR wins, not significant). The paper reports -0.57 . This minor discrepancy likely arises because the “ablated” model has a slightly different implementation than the Separate GARCH benchmark (same parameterization but different initial conditions or refit schedule). The qualitative conclusion is identical: removing the cross-session link makes the model statistically indistinguishable from GJR. Acceptable.

Table 4: VaR and ES

Row	Paper VR	JSON source	JSON VR	Status
TAIFEX PRG Ext	0.95%	K874d PRG Basic	0.949%	Label error (M1)
TAIFEX GJR	2.49%	K874d GJR	2.491%	OK
SPY PRG Ext	0.93%	K880 PRG Ext	0.933%	OK
SPY GJR	1.92%	K880 GJR	1.920%	OK
QQQ PRG Ext	1.26%	K881 QQQ PRG Ext	1.262%	OK
GLD PRG Ext	1.12%	K881 GLD PRG Ext	1.116%	OK
EEM PRG Ext	0.40%	K881 EEM PRG Ext	0.404%	Over-conservative (M7)

6. Specific Issue List

Severe Issues (Must Fix)

S1 Body text “−4.15” inconsistent with Table 2 “−4.07” for TAIEX PRG-vs-Sep.

Section 4.1 (line 195): “DM statistics range from −4.15 (TAIFEX) to −6.69 (SPY)”

Table 2 (correct): −4.07

K874c experiment (correct): $t = -4.0657$

This is an incomplete revision: the table was updated but the body text was not. The same number should also be checked in the abstract (not currently there, so no issue).

Fix: Change −4.15 to −4.07 in line 195. Also change the range in line 295 from “−5.16 to −6.69” to “−4.07 to −6.69” (or explicitly state which markets are included).

Medium Issues (Should Fix)

M1 Table 4 TAIEX VaR row still uses PRG Basic numbers labeled as PRG Extended.

Carried from Round 1 M1. K874d confirms: the reported values (VR = 0.95%, Kupiec $p = 0.88$) match PRG Basic, not PRG Extended (which has VR = 0.24%, Kupiec $p = 0.008$). Either relabel or use correct PRG Extended numbers. If PRG Extended is too conservative for TAIEX VaR, this should be discussed, not hidden.

M2 EEM VaR over-conservatism not acknowledged.

Carried from Round 1 M7. PRG Extended on EEM: VR = 0.40% (1% VaR) and 3.46% (5% VaR), both rejected by Kupiec. The “consistent ranking” claim in Section 4.3 should note this exception.

M3 Acerbi–Szekely (2014) mentioned by name but not formally cited.

Section 4.3 references “the Acerbi–Szekely backtest” without a cite command or bibliography entry. Add:

```
bibitem[Acerbi and Szekely(2014)]{Acerbi2014}
Acerbi, C. and Szekely, B. (2014).
Back-testing expected shortfall.
Risk, 27(11), 76–81.
```

M4 Discussion section PRG-vs-Sep range “−5.16 to −6.69” omits TAIEX and 0050.TW.

Table 2 shows the full range is −4.07 to −6.69. The current text gives the impression that −5.16 (GLD) is the weakest result, when it is actually −4.07 (TAIFEX). Fix: “−4.07 to −6.69” or explicitly restrict to “U.S. ETFs (−5.16 to −6.69).”

Minor Issues (Optional Fix)

m1 Log-likelihood function. Carried from Round 1 M4. Adding the one-line Gaussian quasi-log-likelihood formula ($\ell = \sum_n [-\frac{1}{2} \log h_n - r_n^2 / (2h_n)]$ plus constant) would cost one line and improve reproducibility. Not strictly required for FRL but recommended.

m2 Contribution 2 framing. Carried from Round 1 M3. The introduction still says “develops a fair comparison framework” which is slightly overstated since the framework itself is Patton (2011) / Hansen and Lunde (2005), and the key finding (HAR≈GJR on common target) is

specific to TAIFEX. The body text handles this well; a minor softening in the introduction (e.g., “applies” instead of “develops”) would align the framing.

- m3 ρ dual usage.** Carried from Round 1 m2. ρ_s denotes persistence while Spearman ρ appears in Table 2. Consider ρ^{Sp} for the latter.
- m4 Overnight–intraday return correlation.** The additivity assumption $\sigma_{full}^2 = r_{d,0}^2 + r_{d,1}^2$ relies on zero correlation between session returns. Reporting the sample correlation (expected to be near zero) would reassure readers.
- m5 Transaction costs in VT strategy.** Section 4.4 acknowledges this limitation but draws strong conclusions (“64% improvement”). Even a rough adjustment ($2 \text{ bps} \times \text{average turnover}$) would be helpful.

7. Summary and Recommendations

Overall assessment: The paper has improved substantially from Round 1. All three original severe issues are resolved in the tables and bibliography. The model specification is now complete (Gaussian QMLE, Separate GJR clarified), and the bibliography is clean. The core results are strong and fully verified against experiment data.

What remains:

1. **[S1, 5 minutes]** Fix the stale -4.15 in line 195 and the -5.16 range in line 295. These are simple find-and-replace corrections.
2. **[M1, 15 minutes]** Correct or relabel the TAIFEX VaR row in Table 4. If PRG Extended is too conservative, acknowledge it.
3. **[M2, 5 minutes]** Add a footnote about EEM VaR over-conservatism, or soften the “consistent ranking” claim.
4. **[M3, 5 minutes]** Add the Acerbi–Szekely (2014) bibitem.
5. **[M4, 2 minutes]** Fix the range in Discussion to match Table 2.

After fixing S1 and M1–M4, the paper is ready for submission to Finance Research Letters.

Estimated revision time: 30 minutes.

Key strength (unchanged from Round 1): The ablation study remains the paper’s killer argument. The 6.57σ gap between full PRG and ablated PRG is clean, convincing, and difficult for reviewers to dismiss. The cross-market evidence (6 markets, all DM $|t| > 3.0$) is unusually comprehensive for an FRL paper.